

EU ETS Strategy for Fleet Operators

Financial Exposure Mitigation, European Union Allowances (EUA) Management, and Operational Tuning

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1. Introduction to Maritime EU ETS

The inclusion of maritime transport in the European Union Emissions Trading System (EU ETS) imposes a direct carbon cost on commercial voyages calling at EU/EEA ports. Shipowners and ISM Managers are obligated to surrender European Union Allowances (EUA) for 100% of emissions on intra-EU voyages and 50% of emissions on extra-EU voyages arriving at or departing from EU ports.

100%

FULL PHASE-IN COMPLIANCE

100% / 50%

INTRA-EU / EXTRA-EU SCOPE

€70 - €90

EST. EUA PRICE / TON CO₂

2. Financial Impact Assessment on Capesize / Supramax Fleet

For a standard 58,000 DWT Supramax bulk carrier consuming 22 MT VLSFO/day on a 14-day voyage from Brazil to Rotterdam:

- **Total CO₂ Emissions:** ~1,000 MT CO₂
- **Surrender Obligation (50% Scope):** 500 EUA credits
- **Direct Carbon Cost:** ~€40,000 per single voyage at current EUA market prices

3. Mitigation Strategies

3.1 Technical Retrofits

Deploying fast-payback energy-saving retrofits (Mewis Ducts, VFDs, Foul-Release Paint) reduces fuel burn by 10%–15%, yielding direct cash savings on required EUA purchases equal to €5,000–€8,000 per transatlantic voyage.

3.2 Drop-in Biofuel Blends (B30 / HVO)

Utilizing sustainable biofuels reduces the fossil CO₂ conversion factor (C_f), drastically lowering the required EUA surrender volume while maintaining compliance with FuelEU Maritime intensity targets.

Fuel Type	C _f Factor (t-CO ₂ / t-Fuel)	EUA Obligation Ratio	FuelEU GHG Impact
VLSFO / MGO	3.114 / 3.206	100% (Baseline)	Penalty Zone
Biofuel B30 Blend	2.180 (Well-to-Wake Adj)	~70% Baseline	Compliant
100% HVO Bio-MGO	0.000 (Fossil Equivalent)	~0% Fossil EUA	High Surplus Credit

Strategic Recommendation: Combine operational speed optimization (Just-In-Time port arrival) with physical technical retrofits to minimize exposure to EU carbon tariffs.